Rigid Body Motion (Classical Mechanics)

We are combining N - particle to create 1 ~~p~~ system.

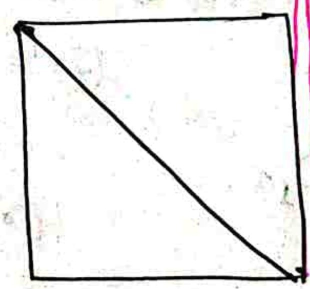
- Rigid body is a system of N - particles in which, distance between any/~~of~~ all two particles remain constant.

$\boxed{r_{ij} = C_{ij}}$ ⇒ r_{ij} is distance from i^{th} particle to the j^{th} particle → constant of motion (imposed one)

Hence, it is a constraint

Distance constraint

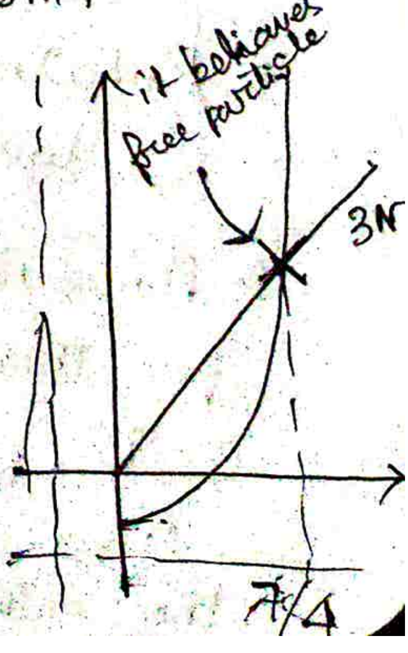
$N^2 \quad r_{ij} = r_{ji}$



$\frac{N(N-1)}{2}$ → no. of independent ~~constraints~~ pairwise ~~coming in~~ distances among n - particles.

- N particles: 3N degrees of freedom.

	$\frac{N(N-1)}{2}$	If free particle
1	0	3
2	1	6
3	3	9
4	6	12
5	10	18



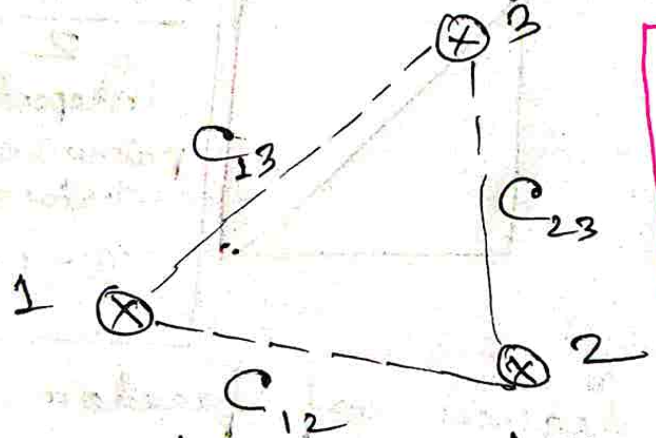
2D

2d • Let us start with particle 1, which is free to move. But particle 2 has to move in a circle of fixed radius (distance) C_{12} from the 2nd particle.

⇒ Position of 1st particle and an angle can completely describe the system.

→ In this frame, we are left with only 1 degree of freedom, that is, rotation about a fixed point.

• Now, we introduce particle 3:



The third particle position remains fixed w.r.t. to positions of particle 1 and 2 given.

• Rotational degree of freedom of particle 2 does not exist anymore.

▷ However, a reflection symmetry exists for this particular 3-particle case in 2-D.

Once we fix 3 points, we essentially have the entire system described.

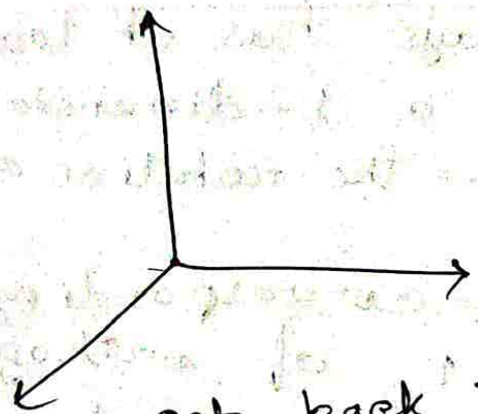
- All non-planar rigid bodies require 6-degrees of freedom in 3 dimensions.

↳ 3 degrees out of these are the coordinates of any 1 point of the rigid body.

→ other 3 degrees are angles, rotations about that point.

▷ Which 3 rotations?

Ans.



$$\vec{X} = \begin{pmatrix} X \\ Y \\ Z \end{pmatrix}$$

▫ If we perform 3 independent rotations, we get back the original ~~syste~~ system.

▫ Rotation is represented by $X' = AX$.

$$\Rightarrow \boxed{AA^T = A^T A = \mathbb{I}}$$

▫ Rotation about z-axis:

$$\boxed{\vec{\xi} = DX}, \quad \boxed{D^T D = \mathbb{I}}, \quad \boxed{\hat{z} = \begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix}}$$

Rotational matrix $D\hat{z} = \hat{z}$ [$\hat{z}$ is an eigenvector of D with eigenvalue 1]

$$\hookrightarrow \begin{pmatrix} \cos\phi & \sin\phi & 0 \\ -\sin\phi & \cos\phi & 0 \\ 0 & 0 & 1 \end{pmatrix}$$

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3 eigenvectors for 3-D → 3 axes of rotations are possible.

- eigenvalue equation will be cubic, hence, all eigenvalues might not be real.

- Special cases, eigenvalues ~~is~~ are real. Eg: Orthogonal matrices.

- Idea of axes of rotation only makes sense when the dimension is odd.

Odd dimensions always has at least one fixed direction i.e. a 1-dimensional invariant subspace — the rotation axis.

⇒ The eigenvector corresponding to real eigenvalue $+1$ of orthogonal matrix $\mathbb{R}^{2N+1}$ is the axis of rotation.

- Suppose; $R(\theta) = \begin{pmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{pmatrix}$

- has eigenvalues $e^{\pm i\theta}$ → no real eigenvector → no axis!

- Now, $\boxed{\xi' = C\,\vec{\xi}}$ about new x-axis by an angle θ.

$$C = \begin{pmatrix} 1 & 0 & 0 \\ 0 & \cos\theta & \sin\theta \\ 0 & -\sin\theta & \cos\theta \end{pmatrix}$$

- $X' = B\xi'$ about new z-axis by an angle Ψ.

$$\vec{B} = \begin{pmatrix} \cos\Psi & \sin\Psi & 0 \\ -\sin\Psi & \cos\Psi & 0 \\ 0 & 0 & 1 \end{pmatrix}$$

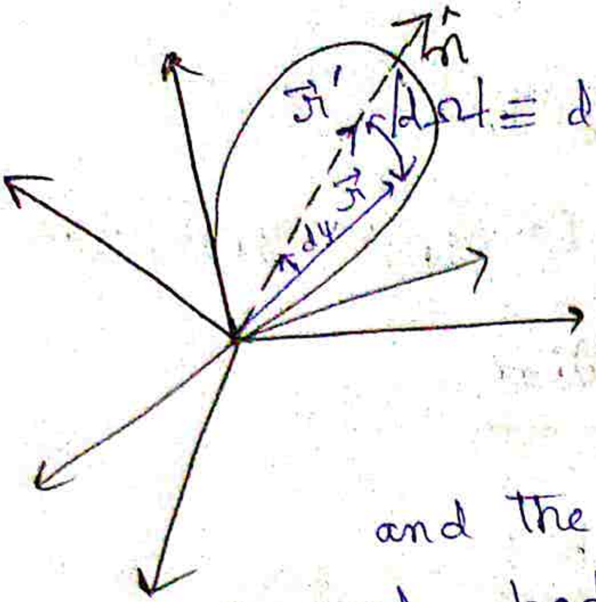
$$\Rightarrow \boxed{X' = AX = BCD\,X} \quad , \quad \boxed{A = BCD}$$

↳ consecutive rotation matrix is also a rotation.

$\boxed{A^T = D^T C^T B^T}$ → Complete set.

$\boxed{DD^T = \mathbb{I}}$, $\boxed{CC^T = \mathbb{I}}$, $\boxed{BB^T = \mathbb{I}}$

⇒ Euler representation of rotation contains full description of rotation in 3-D.



When $t=0$, ~~who~~ there is a fixed frame → inertial frame

$\boxed{\text{spatial frame}}$

As time goes, rigid bodies rotate, and the frame rotates with it. (rigid body not rotating w.r.t. this frame). This frame gives the orientation of the rigid body as a function of time.

Let $\hat{\eta}$ be the axes of rotation. $\Psi \equiv$ angle of rotation.

$$|d\Omega| \to d\Psi \qquad d\vec{\Omega} = \hat{n}\, d\Psi$$

$$\vec{r}\,' = A \cdot \vec{r}\,, \quad A \cdot A^T = \mathbb{1}$$

If $\hat{n} = (0, 0, 1)$

A takes the form: $A = \begin{pmatrix} \cos\Psi & \sin\Psi & 0 \\ -\sin\Psi & \cos\Psi & 0 \\ 0 & 0 & 1 \end{pmatrix}$

- For small rotation; $\Psi \to d\Psi$ $\quad A = \mathbb{1} + \underbrace{\begin{pmatrix} 0 & d\Psi & 0 \\ -d\Psi & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix}}_{\equiv \epsilon}$

$$\Rightarrow A = \begin{pmatrix} 1 & d\Psi & 0 \\ -d\Psi & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}$$

$\epsilon \rightarrow$ antisymmetric matrix.

$$\mathbb{A} = (\mathbb{I} + \epsilon), \text{ Assume elements of } \epsilon_{ij} << 1$$

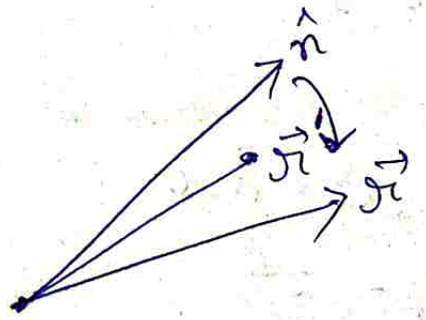
$$\epsilon_{ij}\,\epsilon_{ji} \simeq 0$$

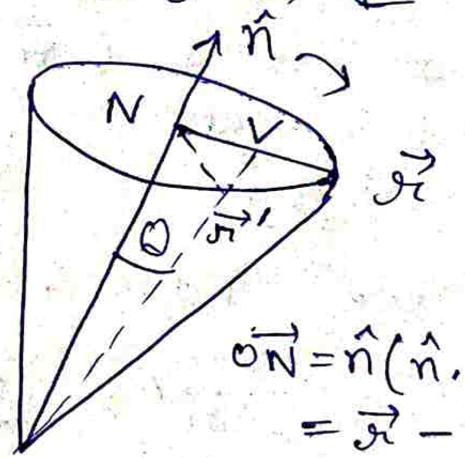
Now, $A \cdot A^T = \mathbb{I} = (\mathbb{I} + \epsilon)(\mathbb{I} + \epsilon)^T$

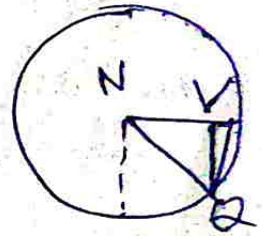
$$= \mathbb{I} + \epsilon + \epsilon^T + \cancel{\epsilon\epsilon^T}^{\,0}.$$

$$\therefore \mathbb{I} + \epsilon + \epsilon^T = \mathbb{1}$$

$$\Rightarrow \epsilon + \epsilon^T = 0 \Rightarrow \epsilon = -\epsilon^T$$







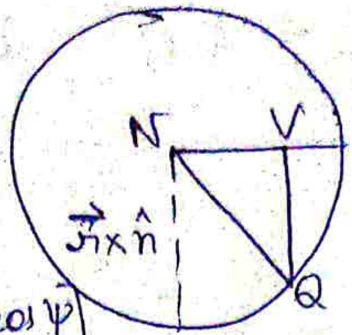
$$\vec{ON} = \hat{n}(\hat{n} \cdot \vec{r}) \quad \vec{NP}$$

$$= \vec{r} - \hat{n}(\hat{n} \cdot \vec{r})$$

$a' =$.

Let us take a vector $\hat{n} \times \vec{r}$

$$\vec{r}\,' = \vec{ON} + \vec{NQ}$$



$$= \vec{ON} + \vec{NQ} + \vec{VQ}$$

$$\vec{NV} = [r - \hat{n}\,(\vec{r}\cdot\hat{n}) \cos\psi]$$

$$VQ = (\vec{r} \times \hat{n}) \sin\Psi$$

$$\vec{r} = \vec{r} \cos\Psi + \hat{n}\,(\hat{n}\cdot\vec{r})\,(1-\cos\Psi) + (\vec{r}\times\hat{n}) \sin\Psi$$

$$\vec{r}\,' = r + (\vec{r}\times\hat{n})\, d\Psi$$

We define a vector:

$$d\vec{\Omega} = d\Psi\,\hat{n}$$

Then we have: $\cancel{\vec{r}\,'} = \vec{r} \times d\Omega$

$$\vec{r}\,' = (\mathbb{I} + \epsilon)\,\vec{r}$$

$$\vec{r} = (x_1, x_2, x_3);\ d\Omega = (d\Omega_1, d\Omega_2, d\Omega_3)$$

$$\vec{r} \times d\Omega = i(x_2\, d\Omega_3 - x_3\, d\Omega_2)$$

$$+ \hat{j}\,(-x_1\, d\Omega_3 + x_2\, d\Omega_1)$$

$$+ \hat{k}\,(x_1\, d\Omega_2 - x_2\, d\Omega_1)$$

$$\begin{pmatrix} 0 & d\Omega_3 & -d\Omega_3 \\ -d\Omega_2 & 0 & d\Omega_1 \\ d\Omega_2 & -d\Omega_1 & 0 \end{pmatrix} \begin{pmatrix} x_1 \\ x_2 \\ x_3 \end{pmatrix} + \begin{pmatrix} x_1 \\ x_2 \\ x_3 \end{pmatrix} =$$

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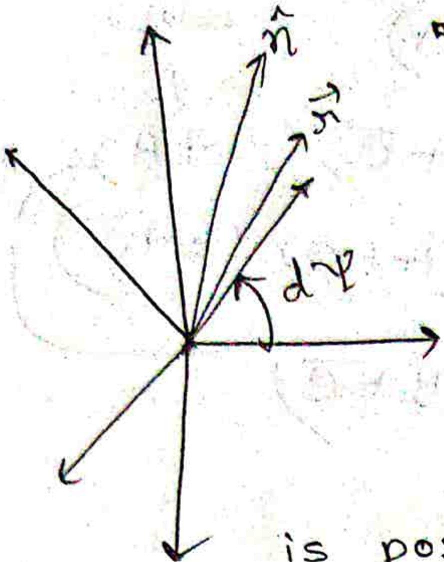


- A frame that is rotating with an arbitrary axis of rotation ($\hat{n}$) and whose origin coincides with the origin of the spatial frame. $\vec{r}$ is position vector w.r.t. the spatial frame. $\vec{r}\,'$ is position vector w.r.t. rotating frame.

- $\vec{r}\,' = \vec{r} + \vec{r} \times \hat{n}\, d\Psi$, $d\vec{\Omega} = \hat{n}\, d\Psi$
- $\vec{r}\,' \simeq \vec{r} + \epsilon \vec{r}$

$$\epsilon = \begin{pmatrix} 0 & -d\Omega_3 & d\Omega_2 \\ d\Omega_2 & 0 & -d\Omega_1 \\ -d\Omega_2 & d\Omega_1 & 0 \end{pmatrix}$$

- This rotation is to bring in the anti-symmetric rotation matrix.

- $(\vec{r} \times \hat{n})_i = \epsilon_{ijk}\, r_j n_k$
- $(\hat{n} \times \vec{r})_i = \epsilon_{ijk}\, n_j r_k\, d\Psi$

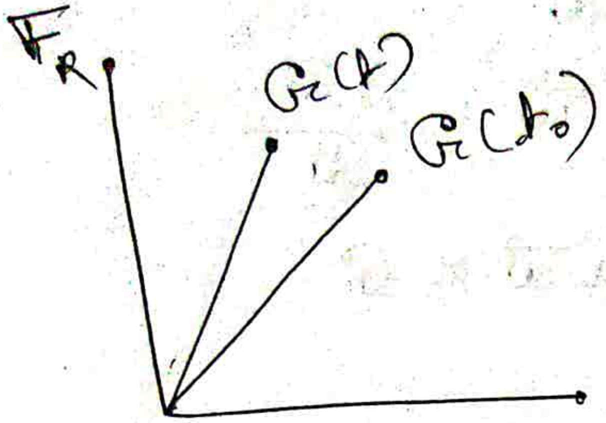
$$= \epsilon_{ijk}\, d\Omega_j\, r_k$$

$d\Omega_i = d\Psi n_i$

$$= \epsilon_{kij}\, d\Omega_j\, r_k$$

$d\Omega_i = d\Psi n_i$, $\epsilon_{kij}\, d\Omega_j = \epsilon_{ki}$

Let G be an arbitrary vector. We see it w.r.t. to the two frames.



($F_R \equiv$ Rotating Frame attached to earth)

If I am ~~on~~ moving on earth, my position is given by $\vec{G}$. But if I am moving around, $\vec{G}$ changes as a function of time.

$$G_s = G_R + dG_{rot}$$

$$d\vec{G} = d\vec{\Omega} \times \vec{G}$$

- When the vector is transformed, the vector looks different due to the transformation.

$$dG_s = G_s(t) - G(t_0) = G_R(t) - G_R(t_0)$$

$$dG_s = dG_R + dG_{rot}$$

$$dG_s = dG_R + \underline{d\Omega \times G}$$

$\hookrightarrow$ How earth is rotating w.r.t. to that rotating frame [When I am fixed w.r.t. to the inertial frame]

$$\lim_{t \to 0} \frac{d r_s}{dt} = \frac{d r_R}{dt} + \frac{d\Omega}{dt} \times r$$

$$\Rightarrow V_s = V_R + \vec{\omega} \times \vec{r} \Rightarrow \vec{\omega} = \frac{d\vec{\Omega}}{dt}$$

- $$\left.\frac{dG}{dt}\right|_s = \left.\frac{dG}{dt}\right|_r + \vec{\omega} \times G .$$

- $$\frac{dV_s}{dt} = \left.\frac{dV_s}{dt}\right|_r + \vec{\omega} \times \vec{V}_s$$

$$\frac{dV_s}{dt} = \frac{d}{dt}\left(\vec{V}_R + \vec{\omega} \times \vec{r}\right) + \vec{\omega} \times \left(\vec{V}_R + \vec{\omega} \times \vec{r}\right)$$

$$= \frac{d\vec{V}_R}{dt} + \vec{\omega} \times \vec{r} + 2\vec{\omega} \times \vec{V}_R + \vec{\omega} \times \vec{\omega} \times \vec{r}$$

$$\Rightarrow a_o = a_R + \vec{\omega} \times \vec{r} + 2\vec{\omega} \times \vec{V}_R + \vec{\omega} \times \vec{\omega} \times \vec{r}$$

The moment we take w to be constant, the whole thing becomes pointless (Why?)

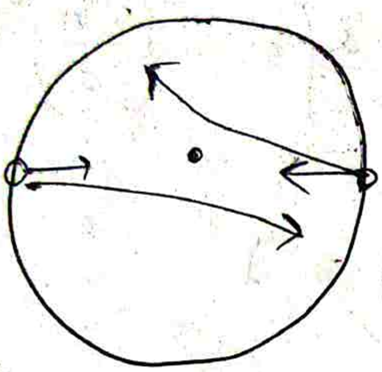
→ Here, $\vec{\omega} \times \vec{r} \longrightarrow$ Euler Force

$2\vec{\omega} \times \vec{V}_R \longrightarrow$ Coriolis Force

$\vec{\omega} \times \vec{\omega} \times \vec{r} \rightarrow$ Centrifugal Force.

- $dG = d\Omega \times G \equiv \epsilon G$

- If ω = constant, coriolis force disappears.
- In north and south hemispheres, velocities in opposite directions.



• → axis of rotation.

If ~~we~~ the two people were sitting diametrically opposite positions on a merry-go-round and spray water at each other, it will get deflected and will not hit the other person.